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DATE: November 12, 2026

SUBJECT: Submission of Regular Research Article: "Anytime-Valid Recalibration of Symmetric Quantum Learners from Free Symmetry Syndromes under Hardware Drift"

Dear Editors of *Quantum*,

We are pleased to submit our original research manuscript entitled **"Anytime-Valid Recalibration of Symmetric Quantum Learners from Free Symmetry Syndromes under Hardware Drift"** for consideration as a Regular Research Article in *Quantum*.

Physical hardware drift—manifesting as telegraphic two-level system (TLS) defect frequency jumps, stochastic local-oscillator phase drift, and energy relaxation decays—presents a major bottleneck for quantum processing units (QPUs) deployed on commercial clouds. Current industry practice relies on rigid, periodic recalibration schedules (typically every 8 to 24 hours), causing up to 80% budget wastage on stable days and silent algorithmic failure during mid-cycle defect jumps. Meanwhile, contemporary drift benchmarking proposals (such as spectator qubits or randomized benchmarking interspersal) require dedicated quantum circuits, consuming 25% to 50% extra quantum shots.

In this manuscript, we present **SymSentinel**, a framework that completely eliminates the shot overhead of quantum hardware monitoring while providing rigorous mathematical guarantees:

- **Zero Extra Quantum Shot Cost ("Free Sentinels"):** We prove that for any symmetric quantum learner ($[U(heta), S] = 0$), computational basis shots gathered for task observable

evaluation inherently carry hardware drift syndromes. These syndromes are evaluated classically in microseconds at **zero extra quantum shot cost**.

- **Filtration-Predictable Betting E-Detectors (Theorem 1):** We construct a sequential test martingale with strictly predictable betting fractions ($\lambda_t \in \mathcal{F}_{t-1}$) guaranteeing non-asymptotic Average Run Length control ($\mathrm{ARL} \ge \gamma$) via Ville's maximal inequality. Over 1,000 independent null runs, our detector achieved **zero false alarms**.
- **The Blind-Spot Visibility Limit Theorem (Theorem 2):** We prove that a single symmetry sentinel S is strictly blind to noise channels whose Kraus operators commute with S ($[E_k, S] = 0$), even when task prediction risk degrades severely. We resolve this via a greedy set-cover algorithm over a sentinel library that restores **100% full Pauli fault visibility** with just $M=2$ sentinels.
- **Anytime-Valid Conformal Risk Certification (Theorem 3):** Combining an empirical-Bernstein confidence sequence on validation data with an isotonic link function, we construct an upper risk certificate U_t^{cert} guaranteeing time-uniform coverage (≤ 0.05 miscoverage) across non-stationary drift regimes (D_1 coherent OU, D_2 relaxation, and D_3 TLS jumps).
- **Three-Tier Recalibration Hierarchy & 96.6% Cloud Cost Reduction (Theorem 4):** Rather than triggering monolithic $100 full QPU resets upon every alarm, SymSentinel deploys a 3-tier cascade: software Virtual-Z phase updates ($0, 0 shots) $ o$ localized 2-qubit subsystem retuning ($15, 1.5k shots) $ o$ full QPU reset. In comprehensive benchmarking across 6 baselines, SymSentinel achieves **96.6% cost reduction compared to periodic recalibration and 82.9% reduction compared to an omniscient oracle reset**.

Given *Quantum*'s leadership in quantum error mitigation, quantum machine learning, and reproducible open science, we believe our work will be of immediate interest to your readership.

Suggested Handling Editors:
• **Dr. Johannes Jakob Meyer** (Freie Universität Berlin) — Symmetries in quantum algorithms.
• **Dr. Robin Blume-Kohout** (Sandia National Laboratories) — Quantum hardware drift and characterization.
• **Dr. Earl Campbell** (Riverlane) — Quantum error mitigation and systems engineering.

Suggested Independent Reviewers:
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(On behalf of Musrat Jahan Gungun and Maheli Ahmed)